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ACETYLSALICYLIC ACID (ASPIRIN): THERAPEUTIC USES

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ABSTRACT

The derivatives of bark of willow are effective analgesics. Aspirin is one of them and has been in use for long time. Aspirin was introduced in 1899 and remains the prototype NSAID. This is the commonest form in which salicylic acid is taken. It inhibits prostaglandin synthesis. Aspirin has anti-inflammatory, antipyretic, analgesic, antiplatelet effects. Large doses produce respiratory stimulation and may cause hypoprothrombenemia, it is well absorbed from the stomach and upper intestinal tract. The resulting salicylate ion is inactivated largely by conjugation with glycine. Aspirin is commonly used as analgesic, antipyretic, and to reduce clumping of platelets in myocardial infarction as a prophylactic measure. The survey of literature regarding the untoward effects clarifies that it may produce gastric ulcer, liver damage and nephropathy if used frequently. It may also produce cutaneous eruptions. To avoid untoward effects of this useful drug its rational therapy is recommended.

History:

The medicinal effects of the bark of willow and certain other plants has been known to several cultures for centuries. Bark of willow has been considered to cure fever. The active ingredient in the willow bark was bitter glycoside called salicin, first isolated in pure form in 1829 by Leroux – who also demonstrated its antipyretic effect on hydrolysis, salicin yields glucose and salicylic alcohol. The latter can be converted into salicylic acid either *in vivo* or by chemical manipulations. Sodium salicylate was first used for the treatment of rheumatic fever and as an antipyretic in 1875 and the discovery of its uricosuric effects and of its usefulness in the treatment of gout, soon followed. The enormous success was obtained when Hoffman prepared acetylsalicylic acid. After demonstration of its anti-inflammatory effects this compound was introduced into medicine in 1899 by Dresser under the name of aspirin. The name has been derived from *Spiraca*, the plant species from which salicylic acid was once prepared (Insel 1990).

INTRODUCTION

Aspirin was introduced in 1899 and remains the prototype NSAID. It is the commonest form in which salicylic acid is taken.

Salicylic acid derivatives

Benorylate: It is an ester of aspirin and paracetamol. It is non-ionic and lipid soluble, well absorbed from the gut, less irritant to the stomach and causes less blood loss. On hydrolysis in the liver and plasma paracetamol and aspirin are released.

Diflunisal: It is fluorophenyl derivative of salicylic acid. It is comparatively better than aspirin in the treatment of osteoarthritis. It causes less gastric adverse effects. It causes diarrhea. It has uricosuric effect.

Methyl Salicycylate: It is used as counter irritant agent in liniments. It is very irritant to stomach so it is only used externally.

Mode of Action: It irreversibly inhibits prostaglandin G/H synthase by acylating the active site of the enzyme, so preventing the formation of products including thromboxane, prostacycline and other prostaglandins; recovery of enzyme action requires resynthesis. Acetyl salicylic acid is rapidly hydrolysed to salicylic acid in the plasma. Salicylic acid besides other actions also has anti-inflammatory action. It also exerts important effects on respiration, intermediary metabolism and acid base balance. It is highly irritant to the stomach.

Effects of Aspirin

Anti-inflammatory effect: Aspirin has anti-inflammatory effect.

Analgesic effect: Aspirin is effective against mild pain of somatic origin.

Antiplatelet effect: This effect is due to permanent inactivation of prostaglandin G/H synthase. Antiplatelet effect is readily achieved with low doses.

Antipyretic effect: Aspirin acts in the hypothalamus to place at a lower level to set point of temperature regulation which is controlled by prostaglandin synthesis.

Respiratory stimulation: Aspirin intoxication stimulates respiratory centre directly and indirectly through CO_2 production.

Large doses of aspirin cause hypoprothrombinemia.

Pharmacokinetics: Aspirin (acetylsalicylic acid) is well absorbed from the stomach and upper intestinal tract. Hydrolysis removes the acetyl group, and the resulting salicylate ion is inactivated largely by conjugation with glycine. Steady plasma concentration can be maintained if aspirin is given six hourly by mouth but high dose given repeatedly may accumulate in toxic amounts, tinnitus may occur.

Uses of Aspirin: Aspirin relieves mild to moderate pain of non-visceral origin, e.g. headache, dysmenorrhoea, osteoarthritis, myalgia and painful bony metastases.

Actions of Aspirin: The study was performed to examine the acute effect of the high dose of aspirin on erebral blood flow in a rabbit model and found that aspirin acutely reduced cerebral blood flow by approximately 20% in a rabbit model, it may be perhaps related to inhibitory effect on prostacycline and/or nitric oxide, this result may help to explain the possible increase in ischaemic stroke seen in low risk patients on aspirin therapy (Bednar and Gross 1999). In a short-term endoscopic study produced significantly less gastrointestinal mucosal damage than either ibuprofen 800 mg tds or aspirin 650 mg QDS and was comparable to placebo in this regard (Lanza 1999). According to the study the long-term effects of very low doses were evaluated. The subjects were randomized to receive 10 mg, 81 mg or 325 mg aspirin daily for 3 months. Before administration of aspirin all subjects underwent gastroduodenoscopy and most underwent proctoscopy for assessment of mucosal injury and prostaglandin content. The findings explain aspirin predominant gastric toxicity and question the safety of even 10mg of aspirin daily (Cryer and Feldman 1999). Aspirin and heparin found to prevent hepatic blood stasis and thrombosis induced by the toxic glycoprotein Bolesatine in mice and it was suggested that these antithrombotic drugs may overcome cases of human poisoning with low exposures of this boletus, showing a hypertension probably due to mechanical obstruction which resists normal therapy (Ennamany *et. al.* 1998). The anti-inflammatory drugs delay the progress of Alzheimer's disease. This selective Cox-2 inhibitors may demonstrate new important therapeutic benefits as anticancer agents, as well as in preventing premature labour and perhaps even retarding the progress of Alzheimer's disease (Vane and Botting 1998).

All NSAIDs including high dose aspirin can cause adverse effects during pregnancy related to the inhibition of prostaglandin synthesis, prolongation of gestation and labour, constriction of the ductus arteriosus, persistent fetal circulation, impairment of renal function and bleeding are risk of third trimester to exposure of pregnant women to all inhibitors of cyclooxygenase. Most of these adverse effects can be prevented by discontinuing NSAID's eight weeks prior to delivery (Ostensen and Ramsey 1998).

Gout in the elderly patients can be treated as colchicine is poorly tolerated and is best avoided. Intra articular and systemic corticosteroids are increasingly being used for treating acute gouty flares in aged patients, with medical disorders NSAID therapy is contraindicated. Urate lowering drugs can be given in hyperuricaemia and chronic gouty arthritis. Uricosuric drugs are poorly tolerated and presence of renal impairment in elderly renders these drugs ineffective. All opurinol is the urate-lowering drug of choice but its use in elderly renders serious cutaneous and hypersensitivity reactions (Fam 1998).

Experimental animal studies have demonstrated a protective effect of NSAIDs against intestinal tumorogenesis due to chemical carcinogens. Many epidemiological studies have shown that regular aspirin use is associated with reduction of the risk of colorectal cancer (Akasu and Kakzoe 1998). Aspirin has been found to induce free fatty accumulation in rat (Yoshida *et al.* 1998). A 7 week treatment with the tobacco carcinogen induced 8-10 lung adenomas per AJ mouse and the hypothesis reveals that PGE_2 synthesis and induction of apoptosis contribute to varying degrees to the mechanism of cancer prevention (Castonguay *et al.* 1998). Aspirin and other NSAIDs reduce the incidence of colorectal cancers and related mortality by 30% to 60% as well as the incidence of colonic adenomas and found beneficial effect in chemoprevention of colorectal and esophageal cancers (Singh and Trotman 1998, Morgan 1999).

Aspirin has been shown to increase endothelial resistance to oxidative damage using cultured cells, the study investigates the effect of aspirin on the expression of ferritin, a cytoprotective protein that sequesters slow cytosolic iron (Oberle *et al.* 1998). Chronic salicylate toxicity should be considered as a cause of systemic inflammatory response syndrome in the absence of a source of infection, since survival appears to be dependent on prompt diagnosis and management (Chalasami 1996).

Aspirin applied by patch to the skin in three subjects showed marked hydrolysis to the inactive product, salicylic acid. Aspirin can be delivered transdermally by patch in a dose that suppress platelet cyclooxygenase (Mc Adam *et al.* 1996).

Untowards Effects

Salicylism: Due to high doses of aspirin symptoms as tinnitis and hearing difficulty, dizziness, headache and confusion may be produced.

Allergy: Due to allergy to aspirin patient may develop severe rhinitis, urticaria, angioedema, asthma or shock. Individuals already suffering from recurrent urticaria, polyps or asthma are more susceptible.

Reyes' Syndrome: This syndrome is rare encephalopathy, liver injury may be produced in children recovering from febrile viral infections.

Over dose: Moderate over dose may produce nausea, vomiting, epigastric discomfort, tinnitis, deafness, hypernoea, headache, sweating, pyrexia, hypokalaemia and restlessness.

Large over dose above symptoms may be followed by pulmonary oedema, convulsions and coma with severe dehydration & ketosis.

Metabolic changes due to toxic dose are respiratory alkalosis, rise in blood pH, metabolic acidosis.

Management of high doses is carried on by doing gastric lavage, correction of dehydration, correction of acid base disturbances, removal of salicylate from the body and haemodialysis may be done.

Aspirin is a common and easily available drug, it is not without adverse effects on stomach, superficial gastric erosions may be produced. There may be occult blood loss, the site of bleeding is stomach. To overcome this adverse effect enteric coated tables are available. History of taking aspirin is common in patients having G.I.T. bleeding. NSAIDs associated toxicity of large and small bowel is increasingly recognized in clinical practice. Enteroscopic procedures are frequently used. There have been several manifestations, ulcerations, strictures, colitis, exacerbations of inflammatory bowel disease, aspirin, diclofenac and sulindac, are most commonly associated with this problem (Bjorkman 1998). Aspirin like drugs may cause injuries including ulcers to the gastrointestinal tract by chelation of the divalent and multivalent metallic cautions in the gastrointestinal mucous and mucosa as suggested by the chemical properties and supported by experimental and clinical data (Wang 1998). In a study the histories, radiologic findings and pathologic findings in two men and two women were reviewed all in sixties with a history of long-term nonsteroidal anti-inflammatory drug or aspirin. The aspirin users presented with symptoms of esophageal disease (Zalv 1998). The association of gastrointestinal damage with the use of non-steroidal anti-inflammatory drugs remain the major limitation to their use (Wallace 1997). Low dose aspirin is widely employed as antiplatelet therapy for cardiovascular disorders, the drug maintains its ability to damage the gastric mucosa (Guslandi 1997). According to literature review about pathogenesis gastroenteropathy continues to be a major limitation to the use of NSAIDs (Wallace 1996). Gastric adaptation to acetylsalicylic acid or stress enhances mucosal resistance to the injury induced by strong irritants, and this appears to be mediated by mucosal regeneration, probably resulting from increased luminal and mucosal contents of epidermal growth factor and excessive expression of its receptors, to conclude it the study was done by administration of acidified acetylsalicylic acid repeatedly for four consecutive days (Brzozowski 1996). Aspirin and non-aspirin non-steroidal anti-inflammatory drugs almost invariably cause acute gastrointestinal injury. Healing of NSAID ulcers by conventional doses of H_2 antagonists is slow. These problems can be overcome by the use of more potent acid suppression. Mioprostol heals NSAID associated ulcers, but misoprostol has adverse events. The use of omeprazol or misoprostol or misoprostol should be considered (Hawkey 1996). According to study in which twelve healthy volunteers were given plain aspirin 300 mg, plain aspirin 75 mg, enteric coated aspirin reduces acute gastric mucosal injury to placebo levels, despite its inhibition of prostaglandin synthesis. Enteric coating is appropriate way to prevent gastric mucosal damage (Cole *et al.* 1999). Aspirin was given to rats to assess the liver damage, the laminar structure which is negative for bilirubin staining seems to be one of the most sensitive indicators of liver damage (Tomoda 1998). An article discusses specific NSAIDs available can cause hepatic toxicity. Most non-steroidal anti-inflammatory drugs produce injury in an unpredictable fashion by way of an idiosyncratic (immunologic versus metabolic) mechanism. Acetaminophen and aspirin are more predictable as they cause injury to liver in dose-dependent manner (Fry and Seef 1995).

Three cases have been reported in whom aspirin was medicated for secondary prophylactins of myocardial infarction but in whom a remote history of an untoward reaction to it prevented its initial use out of three, two patients were found not to be sensitive and started aspirin, the third one was having asthmatic reaction who was successfully desensitized to aspirin allowing for its use (Schaefer and Gore 1999). In an experimental study arachidonic acid was injected into the femoral artery caused hind limb gangrenes. The combined administration of aspirin and ticlopidine prevented the progression of arachidonic acid induced hind limb gangrene and these results suggest that plate lets are involved in the progression of arachidonic acid induced hind limb gangrene. This experimental study can be a valuable novel for the treatment of peripheral vascular disease (Tanaka 1999).

Heavy use of analgesics over the counter products have long been associated with chronic renal failure. Several case control studies have reported associations between chronic renal failure including acetaminophen, aspirin ad other NSAIDs (Mclaughlin 1998). Analgesic nephropathy in rodents was revealed due to exposure to aspirin. There occurred renal papillary necrosis with renal morphological and functional changes similar to that described in humans. It remains to be determined what chemical entities cause classic analgesic nephropathy in humans. Rodents are not appropriate models to decide mechanism of analgesic nephropathy in man (Schnellmann 1998). A 50 years old woman without chronic renal insufficiency was admitted for confusion, cerebellar ataxia associated with hyperchloremic acidosis, she was treated with acetazolamide for glaucoma and with aspirin for acute pericarditis. She developed salicylism. Aspirin enhances free form of acetazolamide. It concluded reciprocal augmentation of the toxicity of acetazolamide and aspirin (Hazouard *et al.* 1999). Six salicylic acid derivatives namely acetyl salicylic acid (aspirin), salicylic acid, salicylamide, sodium salicylate, diflunisal and niclosamide were given to mice and only diflunisal and niclosamide showed genotoxicity (Giri 1996).

A 21 year old woman who had been taking several kinds of analgesics to treat dysmenorrhoea developed episodic attacks of purpuric macular eruption and a burning sensation on exposed areas of the upper chest and back where she had sustained severe sunburn eight months earlier. Target like lesions developed on these areas after intake of Bufferine a combination of aspirin and dialuminate. The case was diagnosed as photo-recall-like phenomenon (Lee 1998).

Cutaneous side effects of non-steroidal anti-inflammatory drugs include urticarial reactions, piroxicam can lead to phototoxic or photoallergic dermatitis. Ketoprofen and bufexamac were major contact allergen (Gebhardt 1995).

There is evidence that interstitial nephritis results mainly from a delayed hypersensitivity response to NSAIDs, and nephrotic syndrome results from changes in glomerular permeability mediated by prostaglandins and other hormones (Kleinknecht 1995).

CONCLUSION

Non-steroidal anti-inflammatory drugs (NSAIDs) including acetylsalicylic acid are commonly used as analgesics and antipyretis. People suffering from joint pain (arthritis) regularly use analgesics, they may develop gastric ulcer and also nephropathy. However they are ought to take the drugs. In such situations only rational therapy is stressed.

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